Name: Nelson Puthuraja Nadar **Roll no.:** 21 **Class/Batch:** SE / INFT-B / B-1

Laboratory Exercise

a. Answer the following:

1. Explain Continuous Integration

Continuous Integration (CI) is a software development practice where developers frequently merge code changes into a central repository, triggering automated builds and tests. It enables teams to detect integration issues early, reduces manual work, and ensures a stable codebase. CI is a key component of DevOps, promoting rapid feedback and higher software quality.

Key Aspects of Continuous Integration:

Frequent Merging: Developers commit code to a shared repository (e.g., Git) multiple times a day to avoid large, complex integration conflicts.

Automated Testing & Building: Upon commit, automated systems immediately build the application and run unit tests, integration tests, or quality checks to verify the new code.

Rapid Feedback: If tests fail, the system notifies the team immediately, allowing for quick fixes.

Shared Repository: A single, central repository acts as the source of truth for the project.

2. Why Jenkins is popular. Mention advantages.

Jenkins is popular primarily because it is a free, open-source automation server with an extensive plugin ecosystem and a large, active community, making it highly flexible and adaptable for continuous integration and delivery (CI/CD) pipelines across diverse platforms.

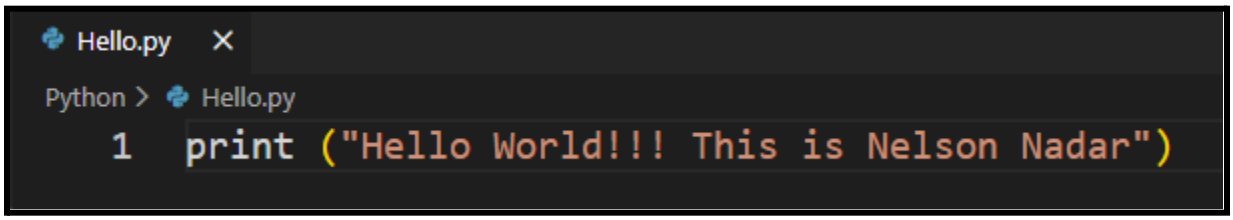
Advantages of Jenkins

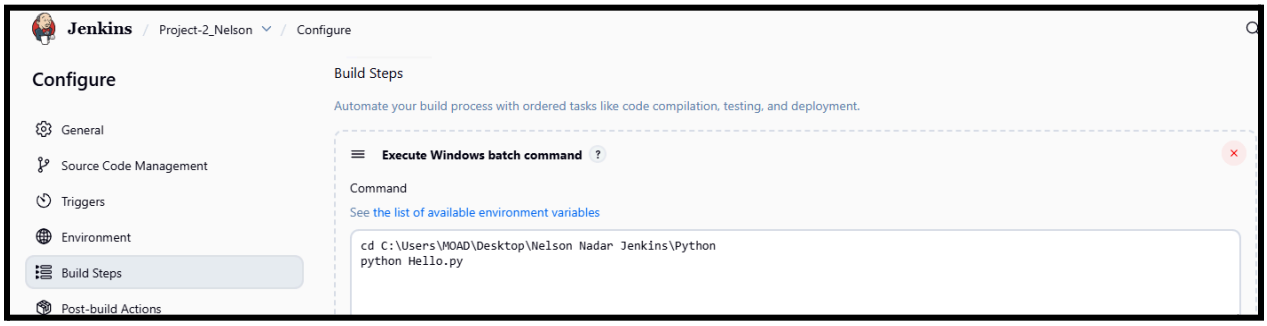
1. Open-Source and Free: Jenkins is a free, open-source tool, which eliminates licensing costs for organizations of all sizes.
2. Vibrant Community Support: It has a large and active community of users and contributors, providing extensive documentation, forums, and a wealth of shared knowledge and contributions.
3. Rich Plugin Ecosystem: With over 1,800+ plugins available, Jenkins can integrate with virtually any tool or service in the software development lifecycle (e.g., Git, Docker, Amazon EC2, Jira), allowing for vast customization and extended functionality.
4. Platform Independence: Built with Java, Jenkins can run on most major operating systems, including Windows, macOS, and Linux.
5. Continuous Integration and Delivery (CI/CD): Jenkins automates the build, test, and deployment processes, enabling continuous integration and continuous delivery, which helps in finding bugs early, reducing manual errors, and accelerating the software delivery lifecycle.
6. Easy Installation and Configuration: Jenkins is a self-contained, Java-based program that is straightforward to install and configure via its web-based graphical user interface (GUI).

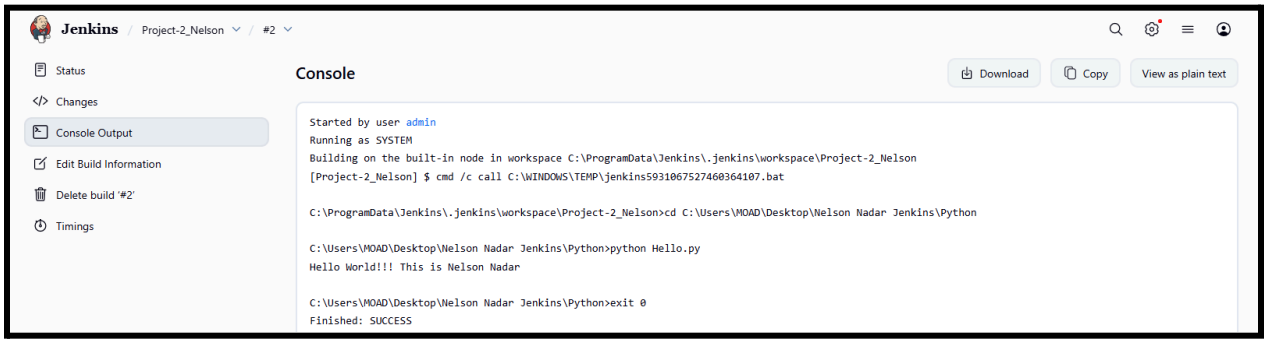
7. Scalability and Distributed Builds: It supports a master-agent architecture, allowing build and testing efforts to be distributed across multiple machines, which reduces build times and handles large workloads efficiently.
8. Code-Based Pipelines: Jenkins supports defining pipelines as code using a Jenkinsfile, which can be stored in a source code management (SCM) repository, enabling versioning and collaboration.
9. Improved Code Quality: Automated testing and continuous feedback on build failures ensure that defects are caught and addressed early, leading to higher overall software quality.

b. Execute following and attach screenshots:

1. Python freestyle project – simple python script (implicit)





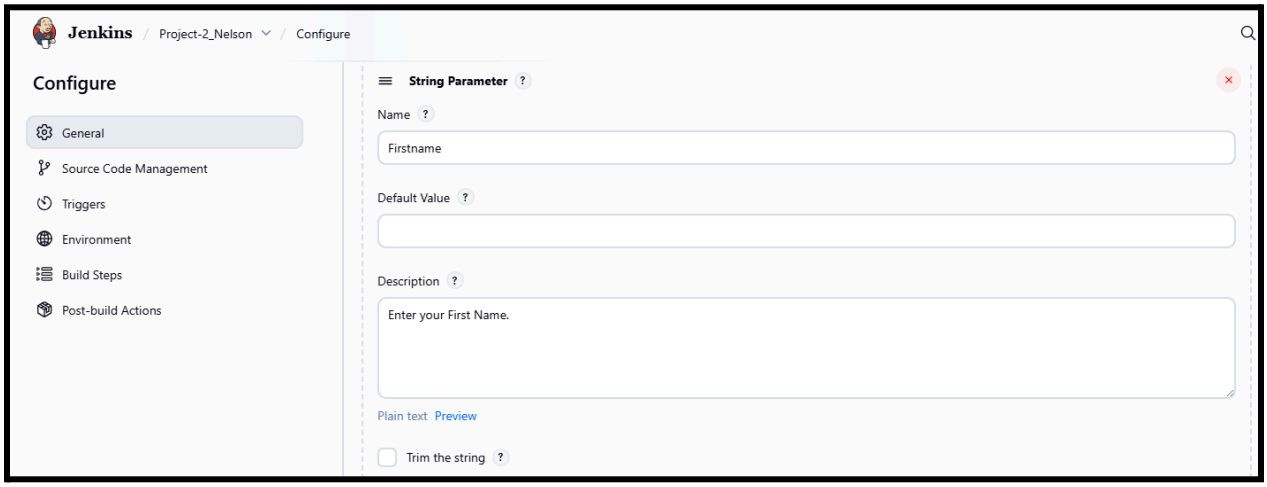


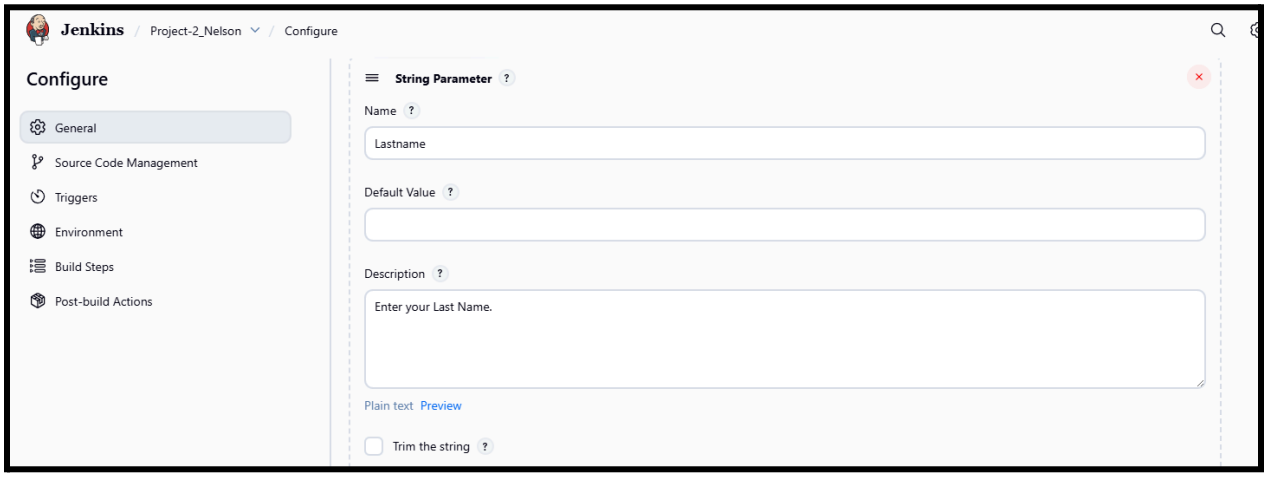
2. Python freestyle project – simple python file (explicit)

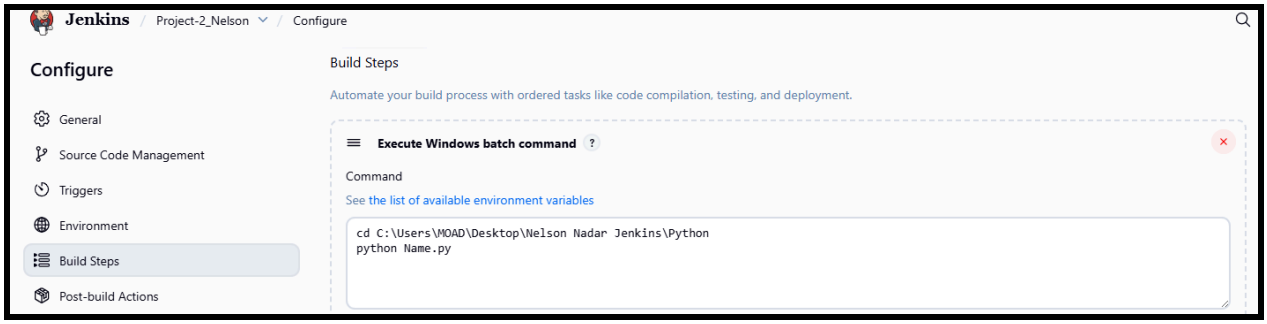
Hello.py Name.py

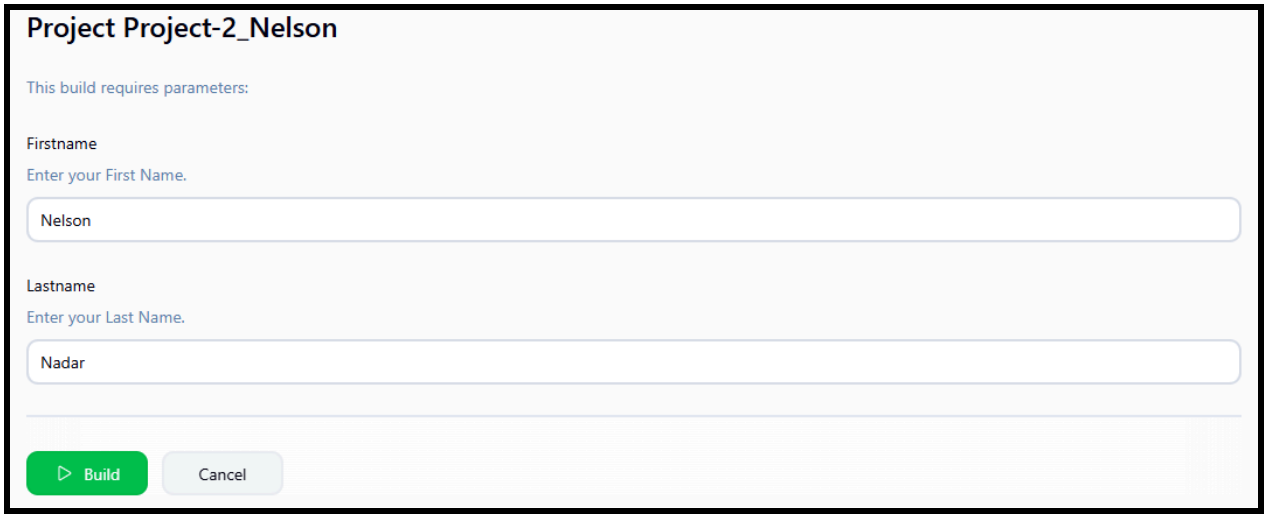
Python > Name.py

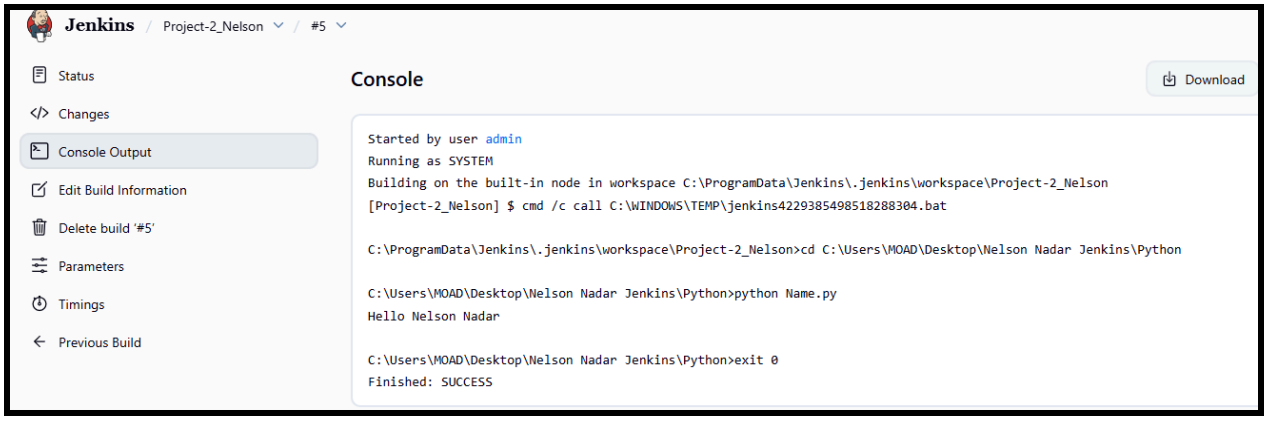
```python
import os
first = os.environ.get("Firstname")
last = os.environ.get("Lastname")
print("Hello", first, last)
```





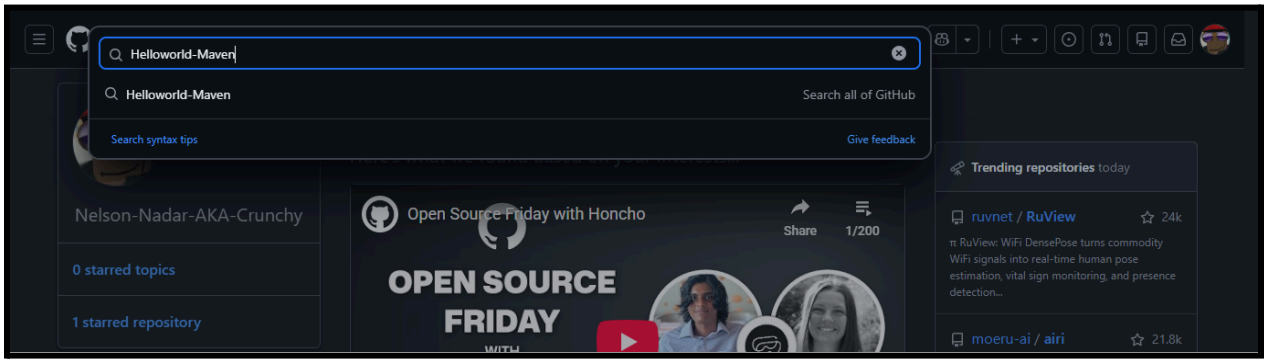


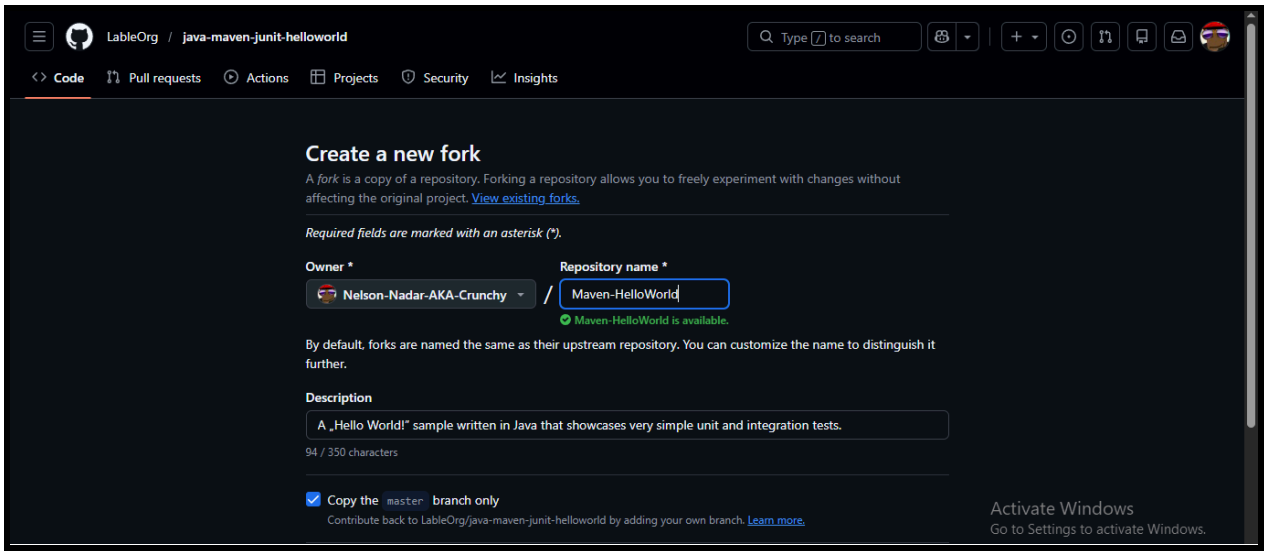


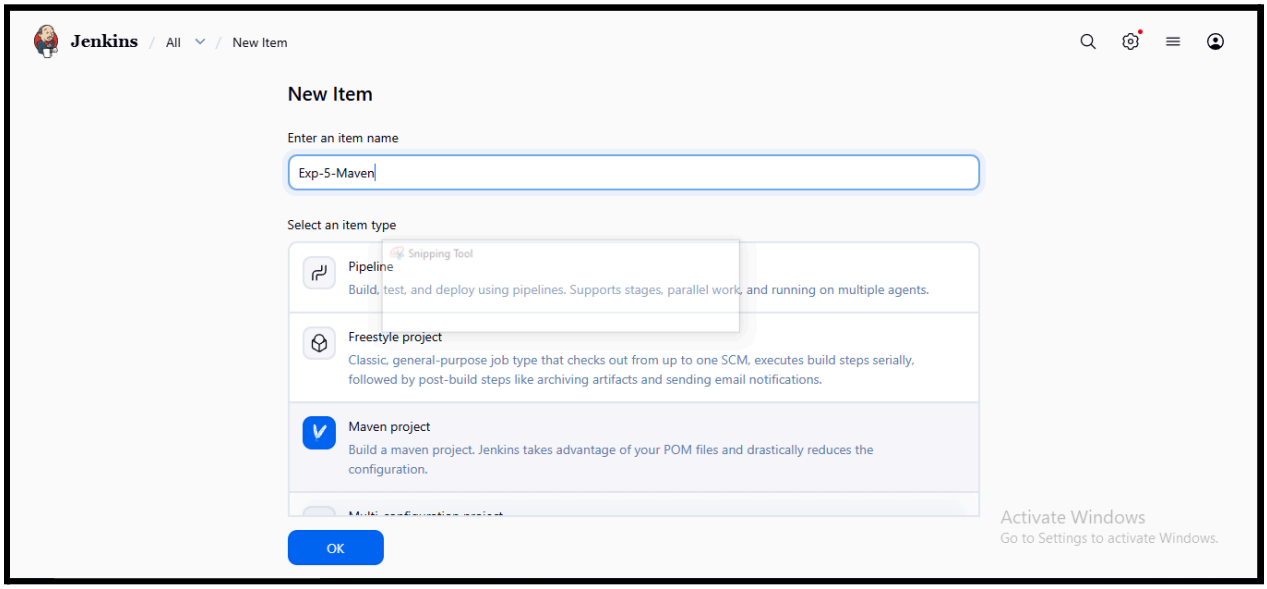


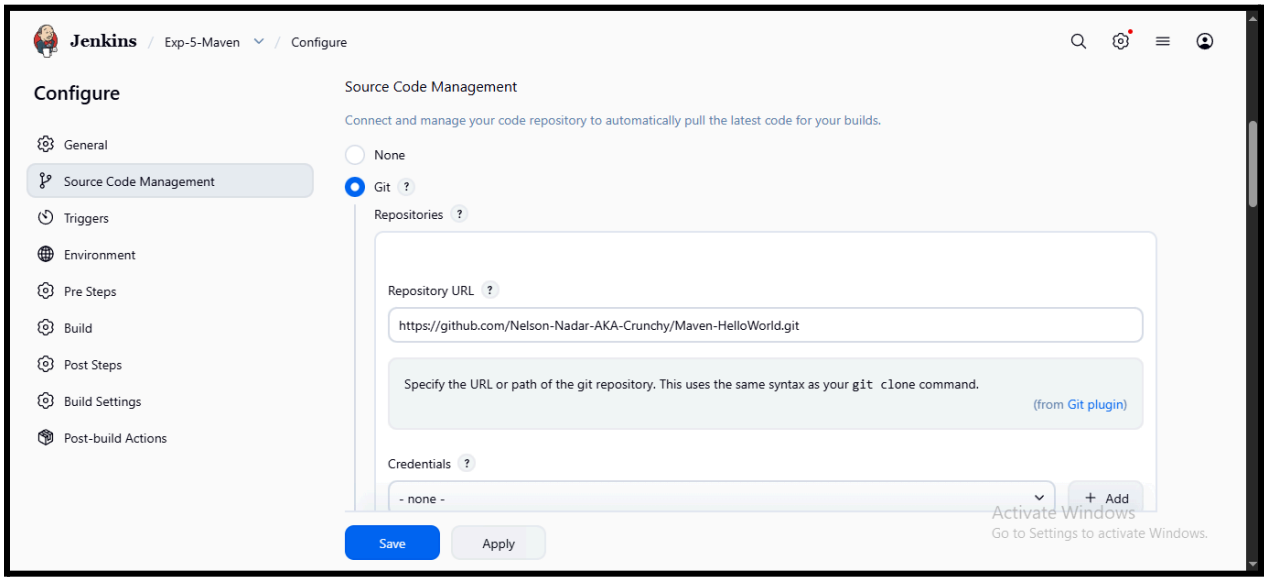
3. Maven freestyle project – Fork repository and Build goals, verify creation of jar/war

Steps:-




LableOrg / java-maven-junit-helloworld
Type / to search
Code
Pull requests
Actions
Projects
Security
Insights
Create a new fork
A fork is a copy of a repository. Forking a repository allows you to freely experiment with changes without affecting the original project. View existing forks.
Required fields are marked with an asterisk (*).
Owner *
Nelson-Nadar-AKA-Crunchy
Repository name *
Maven-HelloWorld
Maven-HelloWorld is available.
By default, forks are named the same as their upstream repository. You can customize the name to distinguish it further.
Description
A „Hello World!" sample written in Java that showcases very simple unit and integration tests.
94 / 350 characters
Copy the master branch only
Contribute back to LableOrg/java-maven-junit-helloworld by adding your own branch. Learn more.
Activate Windows
Go to Settings to activate Windows.


Jenkins / All / New Item
New Item
Enter an item name
Exp-5-Maven
Select an item type
Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
Maven project
Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.
OK
Activate Windows
Go to Settings to activate Windows.


Jenkins / Exp-5-Maven / Configure
Configure
General
Source Code Management
Triggers
Environment
Pre Steps
Build
Post Steps
Build Settings
Post-build Actions
Source Code Management
Connect and manage your code repository to automatically pull the latest code for your builds.
None
Git
Repositories
Repository URL
https://github.com/Nelson-Nadar-AKA-Crunchy/Maven-HelloWorld.git
Specify the URL or path of the git repository. This uses the same syntax as your git clone command.
(from Git plugin)
Credentials
- none -
Add
Save
Apply
Activate Windows
Go to Settings to activate Windows.

Jenkins / Exp-5-Maven / Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Pre Steps
- Build
- Post Steps

Build

Root POM ?

pom.xml

Goals and options ?

clean compile package

Advanced

Jenkins / Exp-5-Maven / #2

- Status
- Changes
- Console Output
- Edit Build Information
- Delete build '#2'
- Timings
- Git Build Data
- Redeploy Artifacts
- See Fingerprints

Console

Download | Copy | View as plain text

```
Started by user Nelson Puthuraja Nadar
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\.jenkins\workspace\Exp-5-Maven
The recommended git tool is: NONE
No credentials specified
 > git.exe rev-parse --resolve-git-dir C:\ProgramData\Jenkins\.jenkins\workspace\Exp-5-Maven\.git # timeout=10
Fetching changes from the remote Git repository
 > git.exe config remote.origin.url https://github.com/Nelson-Nadar-AKA-Crunchy/Maven-HelloWorld.git # timeout=10
Fetching upstream changes from https://github.com/Nelson-Nadar-AKA-Crunchy/Maven-HelloWorld.git
 > git.exe --version # timeout=10
 > git --version # 'git version 2.53.0.windows.1'
 > git.exe fetch --tags --force --progress -- https://github.com/Nelson-Nadar-AKA-Crunchy/Maven-HelloWorld.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
 > git.exe rev-parse "refs/remotes/origin/master^{commit}" # timeout=10
Checking out Revision 9e489de2e27fae2935f4b6a8eed19b90ea4be9eb (refs/remotes/origin/master)
 > git.exe config core.sparsecheckout # timeout=10
 > git.exe checkout -f 9e489de2e27fae2935f4b6a8eed19b90ea4be9eb # timeout=10
Commit message: "Fix Markdown syntax error"
First time build. Skipping changelog.
Parsing POMs
```

Activate Windows
Go to Settings to activate Windows.

Jenkins / Exp-5-Maven / #2

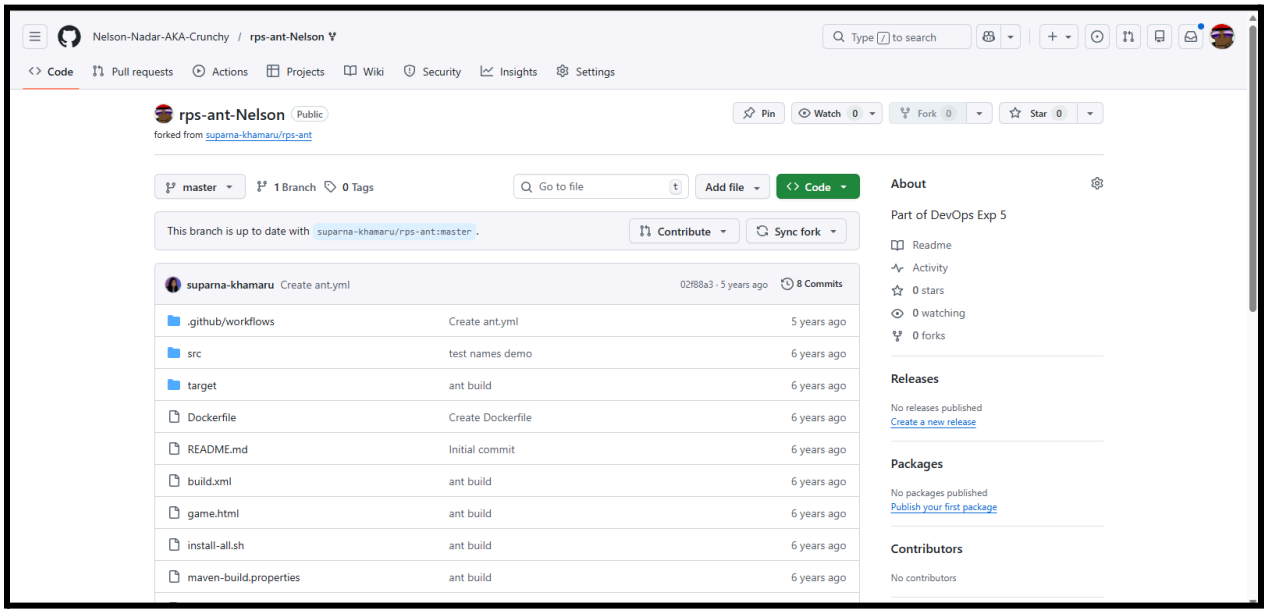
```
at 59 kB/s)
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/com/github/luben/zstd-jni/1.5.5-11/zstd-jni-1.5.5-
11.jar (6.8 MB at 3.4 MB/s)
[INFO] Building jar: C:\ProgramData\Jenkins\.jenkins\workspace\Exp-5-Maven\target\java-maven-junit-helloworld-2.0-
SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  01:27 min
[INFO] Finished at: 2026-03-03T16:24:06+05:30
[INFO] ------------------------------------------------------------------------
Waiting for Jenkins to finish collecting data
[JENKINS] Archiving C:\ProgramData\Jenkins\.jenkins\workspace\Exp-5-Maven\pom.xml to com.example/java-maven-junit-
helloworld/2.0-SNAPSHOT/java-maven-junit-helloworld-2.0-SNAPSHOT.pom
[JENKINS] Archiving C:\ProgramData\Jenkins\.jenkins\workspace\Exp-5-Maven\target\java-maven-junit-helloworld-2.0-
SNAPSHOT.jar to com.example/java-maven-junit-helloworld/2.0-SNAPSHOT/java-maven-junit-helloworld-2.0-SNAPSHOT.jar
channel stopped
Finished: SUCCESS
```

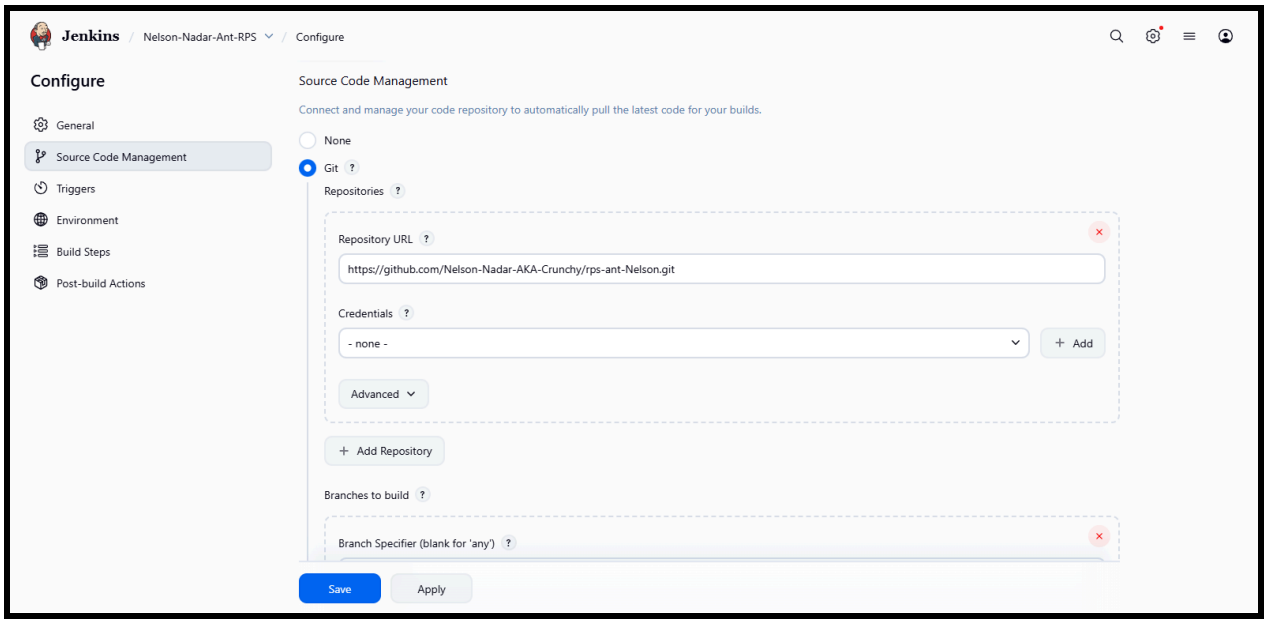
Activate Windows
Go to Settings to activate Windows.

Jenkins 2.550

```
Command Prompt
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Nelson>cd C:\ProgramData\Jenkins\.jenkins\workspace\Exp-5-Maven\target

C:\ProgramData\Jenkins\.jenkins\workspace\Exp-5-Maven\target>dir
 Volume in drive C has no label.
 Volume Serial Number is B650-5E9D

 Directory of C:\ProgramData\Jenkins\.jenkins\workspace\Exp-5-Maven\target

03-03-2026  16:24    <DIR>          .
03-03-2026  16:24    <DIR>          ..
03-03-2026  16:23    <DIR>          classes
03-03-2026  16:23    <DIR>          generated-sources
03-03-2026  16:23    <DIR>          generated-test-sources
03-03-2026  16:23                 0 jacoco.exec
03-03-2026  16:24             5,094 java-maven-junit-helloworld-2.0-SNAPSHOT.jar
03-03-2026  16:24    <DIR>          maven-archiver
03-03-2026  16:23    <DIR>          maven-status
03-03-2026  16:23    <DIR>          surefire-reports
03-03-2026  16:23    <DIR>          test-classes
               2 File(s)          5,094 bytes
               9 Dir(s)  82,011,676,672 bytes free

C:\ProgramData\Jenkins\.jenkins\workspace\Exp-5-Maven\target>
```

4. Ant freestyle project - Fork repository and Build goals, verify creation of jar/war




Jenkins / Nelson-Nadar-Ant-RPS / Configure
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Source Code Management
Connect and manage your code repository to automatically pull the latest code for your builds.
None
Git
Repositories
Repository URL
https://github.com/Nelson-Nadar-AKA-Crunchy/rps-ant-Nelson.git
Credentials
- none -
Add
Advanced
Add Repository
Branches to build
Branch Specifier (blank for 'any')
Save
Apply

Jenkins / Nelson-Nadar-Ant-RPS / Configure
Configure
General
Source Code Management
Triggers
Environment
Build Steps
Post-build Actions
Build Steps
Automate your build process with ordered tasks like code compilation, testing, and deployment.
Invoke Ant
Ant Version
Default
Targets
clean compile package
Advanced
Add build step

Jenkins / Nelson-Nadar-Ant-RPS / #1
Status
Changes
Console Output
Edit Build Information
Delete build '#1'
Timings
Git Build Data
Executed Ant Targets
clean
test-offline
get-deps
compile
test-offline
get-deps
compile
compile-tests
test-junit-present
test-junit-status
junit-missing
test
package
Console
Download
Copy
View as plain text
Started by user admin
Running as SYSTEM
Building on the built-in node in workspace C:\ProgramData\Jenkins\.jenkins\workspace\Nelson-Nadar-Ant-RPS
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/Nelson-Nadar-AKA-Crunchy/rps-ant-Nelson.git
> C:\Program Files\Git\bin\git.exe init C:\ProgramData\Jenkins\.jenkins\workspace\Nelson-Nadar-Ant-RPS # timeout=10
Fetching upstream changes from https://github.com/Nelson-Nadar-AKA-Crunchy/rps-ant-Nelson.git
> C:\Program Files\Git\bin\git.exe --version # timeout=10
> git --version # 'git version 2.47.0.windows.2'
> C:\Program Files\Git\bin\git.exe fetch --tags --force --progress -- https://github.com/Nelson-Nadar-AKA-Crunchy/rps-ant-Nelson.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
> C:\Program Files\Git\bin\git.exe config remote.origin.url https://github.com/Nelson-Nadar-AKA-Crunchy/rps-ant-Nelson.git # timeout=10
> C:\Program Files\Git\bin\git.exe config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> C:\Program Files\Git\bin\git.exe rev-parse "refs/remotes/origin/master^{commit}" # timeout=10
Checking out Revision 02f88a382dda116a84e7f3e5955f827879c04dea (refs/remotes/origin/master)
> C:\Program Files\Git\bin\git.exe config core.sparsecheckout # timeout=10
> C:\Program Files\Git\bin\git.exe checkout -f 02f88a382dda116a84e7f3e5955f827879c04dea # timeout=10
Commit message: "Create ant.yml"
First time build. Skipping changelog.
[Nelson-Nadar-Ant-RPS] $ cmd.exe /C "ant.bat clean compile package && exit %%ERRORLEVEL%%"
Buildfile: C:\ProgramData\Jenkins\.jenkins\workspace\Nelson-Nadar-Ant-RPS\build.xml
clean:
[delete] Deleting directory C:\ProgramData\Jenkins\.jenkins\workspace\Nelson-Nadar-Ant-RPS\target

Jenkins / Nelson-Nadar-Ant-RPS / #1

```
clean:
   [delete] Deleting directory C:\ProgramData\Jenkins\.jenkins\workspace\Nelson-Nadar-Ant-RPS\target

test-offline:

get-deps:
      [get] Getting: https://repo.maven.apache.org/maven2/junit/junit/4.13.1/junit-4.13.1.jar
      [get] To: C:\WINDOWS\system32\config\systemprofile\.m2\repository\junit\junit\4.13.1\junit-4.13.1.jar
      [get] Getting: https://repo.maven.apache.org/maven2/org/hamcrest/hamcrest-core/1.3/hamcrest-core-1.3.jar
      [get] To: C:\WINDOWS\system32\config\systemprofile\.m2\repository\org\hamcrest\hamcrest-core\1.3\hamcrest-core-1.3.jar

compile:
    [mkdir] Created dir: C:\ProgramData\Jenkins\.jenkins\workspace\Nelson-Nadar-Ant-RPS\target\classes
    [javac] C:\ProgramData\Jenkins\.jenkins\workspace\Nelson-Nadar-Ant-RPS\maven-build.xml:74: warning: 'includeantruntime' was not set, defaulting to
build.sysclasspath=last; set to false for repeatable builds
    [javac] Compiling 6 source files to C:\ProgramData\Jenkins\.jenkins\workspace\Nelson-Nadar-Ant-RPS\target\classes
    [javac] warning: [options] bootstrap class path not set in conjunction with -source 8
    [javac] 1 warning

test-offline:

get-deps:
      [get] Getting: https://repo.maven.apache.org/maven2/junit/junit/4.13.1/junit-4.13.1.jar
      [get] To: C:\WINDOWS\system32\config\systemprofile\.m2\repository\junit\junit\4.13.1\junit-4.13.1.jar
      [get] Getting: https://repo.maven.apache.org/maven2/org/hamcrest/hamcrest-core/1.3/hamcrest-core-1.3.jar
      [get] To: C:\WINDOWS\system32\config\systemprofile\.m2\repository\org\hamcrest\hamcrest-core\1.3\hamcrest-core-1.3.jar

compile:
    [javac] C:\ProgramData\Jenkins\.jenkins\workspace\Nelson-Nadar-Ant-RPS\maven-build.xml:74: warning: 'includeantruntime' was not set, defaulting to
build.sysclasspath=last; set to false for repeatable builds
```

Jenkins / Nelson-Nadar-Ant-RPS / #1

```
test:
    [mkdir] Created dir: C:\ProgramData\Jenkins\.jenkins\workspace\Nelson-Nadar-Ant-RPS\target\test-reports
    [junit] Running com.mcnz.FailMeMaybeTest
    [junit] Testsuite: com.mcnz.FailMeMaybeTest
    [junit] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.064 sec
    [junit] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.064 sec
    [junit]
    [junit] Testcase: testMyTrueAssertion took 0.002 sec
    [junit] Testcase: demoTest3 took 0 sec
    [junit] Testcase: demoTest4 took 0 sec
    [junit] Running com.mcnz.rps.GameSummaryTest
    [junit] Testsuite: com.mcnz.rps.GameSummaryTest
    [junit] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.066 sec
    [junit] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.066 sec
    [junit]
    [junit] Testcase: testGetClientGesture took 0.002 sec
    [junit] Testcase: testGetServerGesture took 0 sec
    [junit] Testcase: testGetDate took 0 sec
    [junit] Testcase: demoTest1 took 0 sec
    [junit] Testcase: demoTest2 took 0 sec
    [junit] Testcase: testGetResult took 0 sec

package:
    [mkdir] Created dir: C:\ProgramData\Jenkins\.jenkins\workspace\Nelson-Nadar-Ant-RPS\target\roshambo\WEB-INF\lib
      [war] Building war: C:\ProgramData\Jenkins\.jenkins\workspace\Nelson-Nadar-Ant-RPS\target\roshambo.war

BUILD SUCCESSFUL
Total time: 11 seconds
Finished: SUCCESS
```

```
Command Prompt

Microsoft Windows [Version 10.0.22631.4317]
(c) Microsoft Corporation. All rights reserved.

C:\Users\MOAD>cd C:\ProgramData\Jenkins\.jenkins\workspace\Nelson-Nadar-Ant-RPS

C:\ProgramData\Jenkins\.jenkins\workspace\Nelson-Nadar-Ant-RPS>cd target

C:\ProgramData\Jenkins\.jenkins\workspace\Nelson-Nadar-Ant-RPS\target>dir
 Volume in drive C has no label.
 Volume Serial Number is E6A0-80C9

 Directory of C:\ProgramData\Jenkins\.jenkins\workspace\Nelson-Nadar-Ant-RPS\target

03/10/2026  01:16 PM    <DIR>          .
03/10/2026  01:16 PM    <DIR>          ..
03/10/2026  01:16 PM    <DIR>          classes
03/10/2026  01:16 PM    <DIR>          roshambo
03/10/2026  01:16 PM             9,761 roshambo.war
03/10/2026  01:16 PM    <DIR>          test-classes
03/10/2026  01:16 PM    <DIR>          test-reports
               1 File(s)          9,761 bytes
               6 Dir(s)  124,719,824,896 bytes free

C:\ProgramData\Jenkins\.jenkins\workspace\Nelson-Nadar-Ant-RPS\target>
```

This PC > Local Disk (C:) > ProgramData > Jenkins > .jenkins > workspace > Nelson-Nadar-Ant-RPS > target >

Sort View ...

Name	Date modified	Type	Size
classes	3/10/2026 1:16 PM	File folder	
roshambo	3/10/2026 1:16 PM	File folder	
test-classes	3/10/2026 1:16 PM	File folder	
test-reports	3/10/2026 1:16 PM	File folder	
roshambo.war	3/10/2026 1:16 PM	WAR File	10 KB